MIL-L-52918
AMENDMENT 1
5 January 1981

MILITARY SPECIFICATION

LEVELS, SURVEYING, SELF-LEVELING AND TILTING

This amemdment forms a part of Military Specification MIL-L-52918, dated 3 January 1977, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1, under STANDARDS add:

"FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services."

2.2, under NATIONAL BUREAU OF STANDARDS delete:

"Handbook H28 - Screw-Thread Standards for Federal Services."

PAGE 3

3.1.2, line 2, delete "graduated differential."
3.1.2, end of paragraph, add: Category 1 levels of Type I shall have a graduated differential tilting screw."

After 3.1.3.2, add: "3.1.4 Category 1 levels. First order levels of this category shall be Type I, Style A, with tubular level vials which are rated at 10 arc seconds per 2 mm division or less."

Add: "3.1.5 Category 2 levels. Levels of Category 2 shall be of either Type I or Type II design for second order accuracy. Type I levels of Category 2 shall have tubular level vials rated at 30 arc seconds per 2 mm division or less."

3.2, delete and substitute: "3.2 First article (first-produced level). The contractor shall furnish one or more levels as specified (see 6.2) for examination and testing within the time frame specified (see 6.2), to prove that his production methods and choice of design detail will produce levels that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3)."

FSC 6675

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PAGE 4

3.4, at end of paragraph, add "(see 6.9.)"

PAGE 5

3.7.1(f), delete "7mm" and substitute "6 mm."

PAGES 5 and 6

3.7.1.2, delete and substitute:

"3.7.1.2 Reticle. The reticle shall consist of a pattern of opaque lines permanently registered on optical glass and mounted on a metal reticle cell. The reticle shall be arranged to eliminate distortion during collimation. Means for achieving collimation adjustments shall be positive and accessible. The reticle cell shall be adjustable for precise, positioning of crosslines. Adjustments shall be by means of 3 or 4 capstan head screws that fasten the cell to the telescope and permit rotation of the cell. The reticle cell shall be provided with peg adjustments, a fitted sliding component, and adjustment screws for movement of the reticle in a vertical plane. Category 1 levels may be adjusted by means of an optical wedge at the contractor's options. Horizontal stadia lines shall be not less than 5 minutes of arc in length parallel to the horizontal center line, and spaced to achieve the stadia constant specified (see 6.2). The rod shall be located at a predetermined distance greater than 100 feet or 30 meters from the level. Readings shall not result in errors that exceed 0.3 percent of the horizontal distance and shall not include allowance for the instrument constant. The width of all reticle lines except Category 1 levels shall be 3 seconds of arc plus 0 seconds minus 0.5 seconds. Category 1 level reticle line width shall be 3 microns, plus 0 minus 0.5 microns."

PAGE 6

3.8.1, delete and substitute: "Type I Levels. On Style A levels, the circular level vial shall have a sensitivity of 10 minutes of arc, plus or minus 2 minutes, per 2 mm division. On Style B levels, the level sensitivity shall be 4.5 minutes of arc, plus or minus 1 minute."

3.8.2, line 2, change "7 to 15 minutes" to read "7 to 10 minutes".

3.8.3, line 1, delete "Type I and".

3.9.1, lines 2 and 3 delete second sentence and substitute: "The telescope level vial shall have a sensitivity of 8 to 10 seconds of arc per 2 mm movement for Category 1; Category 2 shall have a level vial sensitivity not more than 30 seconds of arc."

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PAGE 7

3.9.2, line 7, change "not less than ±15" to "not more than plus or minus 15 ..."

3.9.2, line 11, following line 10 insert "The suspension shall insure uniform freedom of".

3.10, line 5, delete "Handbook H28" and substitute "FED-STD-H28."

PAGE 10

4.2(a), delete and substitute "First produced level examination (see 4.3)."

PAGE 11

4.3, delete and substitute "First produced inspection."

PAGES 13 and 14

Table I, first column heading, delete "Qualification" and substitute "First produced".

PAGE 23

6.2, after (h), add: "(i) Time frame required for submission of first produced levels, and numbers of levels required (see 3.2)."

6.3, delete and substitute:

"6.3 First produced levels. Any changes or deviations of production levels from the approved first produced level during production will be subject to the approval of the contracting officer. Approval of the first produced level will not relieve the contractor of his obligation to furnish levels conforming to this specification."

PAGE 24

After 6.8, add:

"6.9 Recycled material. It is encouraged that recycled material be used when practical, provided it meets the requirements of this specification (see 3.4)."

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activity:
DLA - GS

Preparing activity:
Army - ME

Project 6675-0710

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